

Backside Power Delivery With Relaxed Overlay for Backside Patterning Using Extreme Wafer Thinning and Molybdenum-Filled Slit Nano Through Silicon Vias

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Abstract—Backside power delivery network (BSPDN) has gained much attention due to its potential to independently optimize signal and power routing. In this work, long slit nano through silicon vias (nTSVs) is used for high-density connections between frontside (FS)-patterned buried power rails (BPRs) and orthogonally patterned metal rails on the wafer backside (BS). These nTSVs are in situ patterned on top of BPR with self-alignment using FS lithography, and the length of the slits can also be tuned. This design relaxes overlay requirements for BS patterning that are typically stringent due to wafer grid distortions during bonding. Additionally, extreme wafer thinning stopping on a 10 nm $\text{Si}_{0.75}\text{Ge}_{0.25}$ etch stop layer (ESL) is enabled using an optimized thinning sequence with excellent total thickness variation (TTV) control. For the first time, low resistance barrier-free Molybdenum (Mo)-filled nTSVs are demonstrated, confirming the potential for further scaling compared to TiN/W-filled counterparts.

Index Terms—Backside power delivery network (BSPDN), buried power rail (BPR), etch stop layer (ESL), extreme wafer thinning, nano through silicon via (nTSV).

I. INTRODUCTION

WHILE pitch scaling is saturating, technology explorations are shifting toward scaling boosters at standard cell (STC) level or beyond [1], [2]. Among them, the backside power delivery network (BSPDN) has gained many tractions due to the enabled power and performance benefits on top of the flexible STC area reduction [3], [4]. By decoupling

signal and power routing and placing the latter on the wafer backside (BS), the independent optimization for frontside (FS) signal routing and BS power routing can be achieved. A fine-pitch back end of line (BEOL) is implemented exclusively for FS signal wiring. Meanwhile, the relatively relaxed critical dimension (CD) of power wiring on the BS permits a power delivery network (PDN) with reduced resistance. It is also worth noting that the effective chip area usage has been significantly increased.

Various implementation scenarios of BSPDN have been proposed, categorized by the connection schemes between BS and FS wirings, as well as their cointegration compatibility with active devices [5]. Among these scenarios, the buried power rail (BPR)-nano through silicon via (nTSV) last has been extensively investigated [Fig. 1(a)] [6]. In this approach, the FS-embedded BPR is eventually accessed from the BS-defined nTSV after FS processing, wafer bonding, and thinning [7]. The dimensions of nTSVs have been scaled down, and various metals such as copper (Cu, only for micrometer size TSV), tungsten (W), and cobalt (Co) have been demonstrated for nTSV filling. The connection between the nTSV and BPR has evolved from the initial “tap cell” to the more recent “tip-to-tip” connection, which occupies no extra space for the nTSV [6].

However, there are challenges in sustaining these developments. First, to precisely align nTSV on BPR after the bonding, high-order corrections per exposure (HOCPE) are needed to compensate for bonding-induced distortions. In [7], with HOCPE, the overlay between nTSV and BPR was reduced from more than 150 nm to less than 10 nm. However, when the extremely tight overlay is needed, the HOCPE-achieved alignment is expected still to be marginal [8]. Therefore, self-alignment structures defined on the wafer FS should be considered [9]. Additionally, with a maximum aspect ratio of 1:10, the nTSV depth (i.e., remaining wafer thickness) also needs to be reduced along with the CDs to a few 100 nm or even less, which requires a robust extreme wafer thinning approach.

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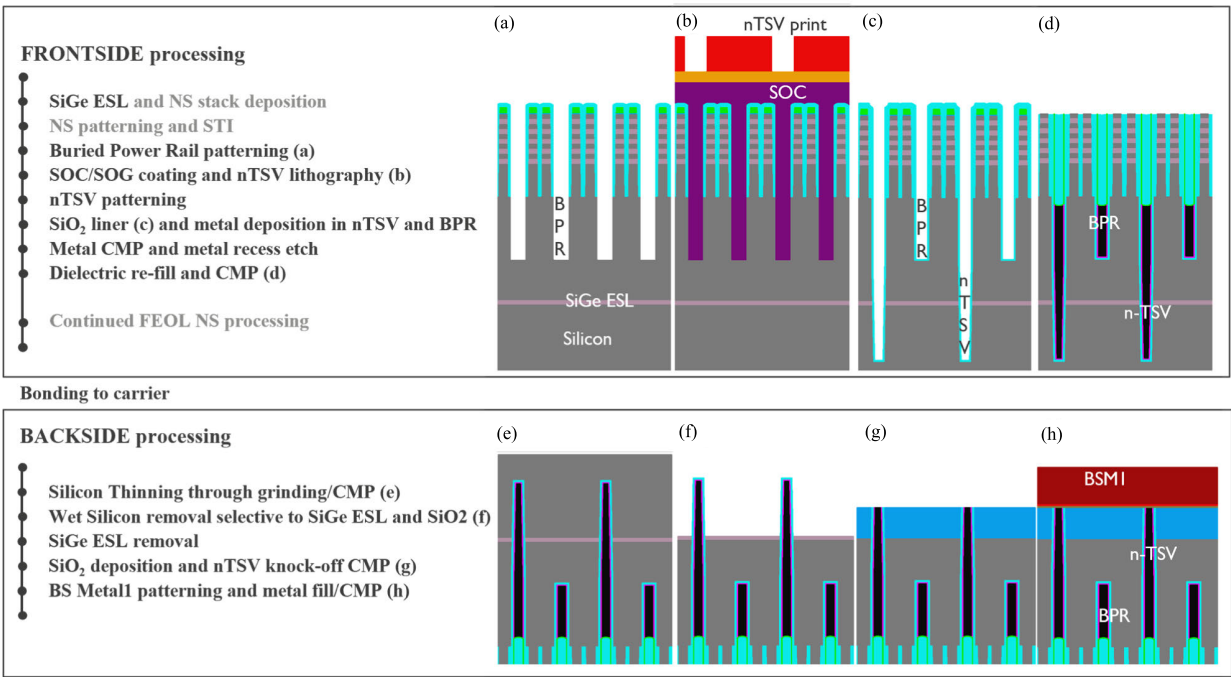


Fig. 2. Process flow and Coventor¹ images. (a)–(d) FS processing. (e)–(h) BS processing. Anticipated processing steps for nanosheet device fabrication are colored in gray.

TABLE I
 COMPARISON OF TWO THINNING SEQUENCE

	SiGe ESL thickness	775 to 50 μ m	50 to 49 μ m	49 to 10 μ m	10 to 3 μ m	3 μ m to SiGe ESL
Previous thinning sequence	50nm	Grinding	Standalone CMP	Dry etching or wet etching I	Wet etching 2	Wet etching 3
Revised thinning sequence	20 or 10nm	Grinding + in-situ CMP				Wet etching 3

However, in the second LC, the TTV average is reduced to less than 1 μ m, with a minimum TTV of 0.4 μ m being achieved (Fig. 3). A typical TTV profile of the revised thinning sequence at approximately 3 μ m Si remaining is shown in Fig. 4(b). In comparison, the TTV profile of the previous thinning sequence is rather hilly [Fig. 4(a)]. Low TTVs achieved from the revised thinning sequence allow the use of a thin SiGe ESL of 20 or 10 nm to compensate for accumulated Si thickness nonuniformity. TEM images post final selective Si wet etching for a wafer with 10 nm SiGe ESL are shown in Fig. 5: the wafer center has 9 nm SiGe remaining, while the wafer edge has 3 nm SiGe remaining. No punchthrough are detected.

B. BS Lithography and Metallization

The BSM1-BPR overlay vector map measured from the scanner during BSM1 lithography is shown in Fig. 6(a). Without the use of HOCPE, the maximum overlay across the wafer including the extreme edge is less than 70 nm, well within

the 100 nm alignment margin (Section I). An inner square with a small overlay can be found at the wafer center. The regions with large overlays are distributed as four symmetric fans around the middle of the wafer. At the same time, the post-bonding residuals vector map is also retrieved [Fig. 6(b)], showing a similar pattern in terms of vector distribution and vector size. This is in line with [14]: the bonding-induced wafer grid distortion is the primary source of BS lithography alignment to FS features. On one hand, HOCPE can be used to compensate for these grid distortions to a certain extent. On the other hand, warpage control measures such as stress compensation layer deposition together with delicately tuned bonding recipes can be implemented to fundamentally reduce bonding distortions.

Since exposed Mo surfaces are prone to be oxidized, special precautions are necessary during the BSM1 Cu metallization process for wafers with Mo-filled vias. First of all, after the knock-off CMP, a thin SiCN layer is deposited on top of exposed vias to protect Mo from oxidation. Following that, once the protection layer is opened, surface cleaning with certain over etch and the deposition of Tantalum (Ta) barrier

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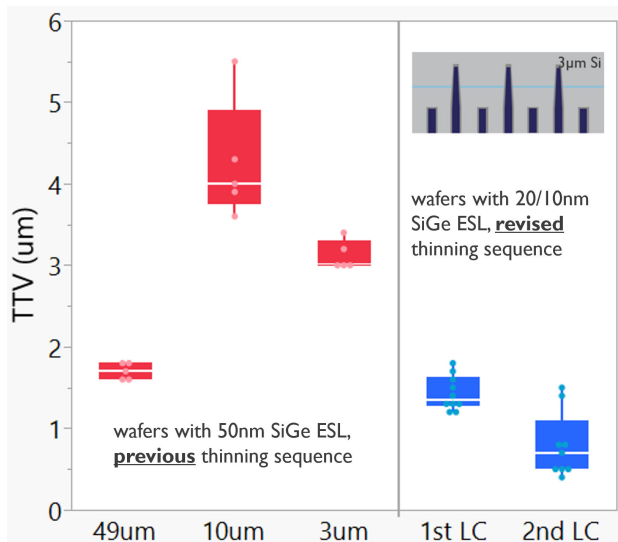


Fig. 3. TTV comparison of different thinning sequences. Data in red are from the previous thinning sequence (as a function of remained Si substrate thickness). Data in blue are from the revised thinning sequence. LC: learning cycle.

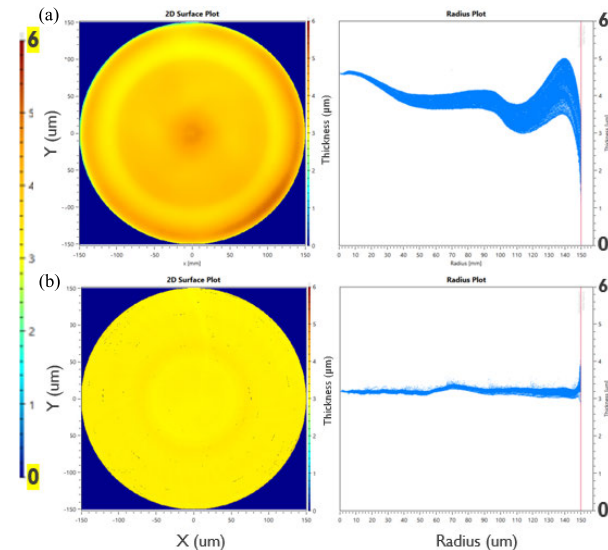


Fig. 4. TTV profile comparison of two different thinning sequences. (a) Previous thinning sequence. (b) Revised thinning sequence. No edge exclusion was implemented for TTV extraction.

deposition is subsequently carried out in a controlled time loop. As shown in Fig. 7(a), TEM with EDS analysis is used to monitor the interface between Mo vias and Cu BSM1, revealing no detectable oxygen signal. For wafers with TiN/W filled vias, although the exposed W vias have limited oxidation risks, a similar EDS analysis is performed, which also confirms an oxygen-free interface [Fig. 7(b)]. After BSM1 metallization and CMP, final TEMs with two orthogonal cuts are shown in Fig. 8, illustrating how the nTSV bridges the FS BPR and BSM1. Cut A is parallel to nTSV and BPR but perpendicular to BSM1, while cut B is the opposite. The 100 nm overlay margin between BSM1 and nTSV is marked accordingly in Cut A.

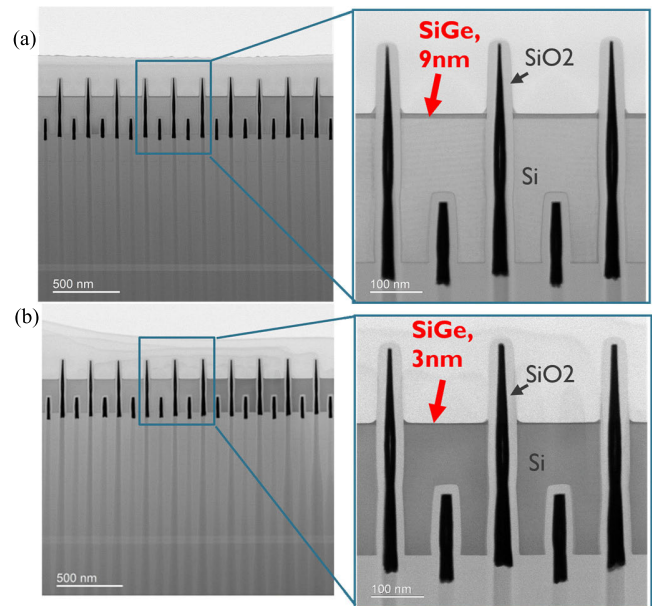


Fig. 5. TEM images post final Si wet etching at (a) wafer center, 9 nm SiGe remained, (b) wafer edge, 3 nm SiGe remained.

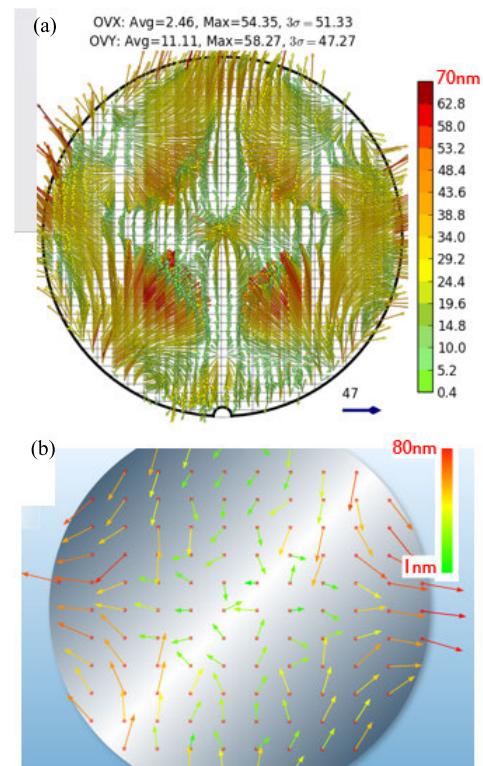


Fig. 6. (a) BSM1-BPR overlay vector map. (b) Residual vector map post bonding.

IV. ELECTRICAL MEASUREMENT AND ANALYSIS

To demonstrate the feasibility of developing nTSV's first integration scheme, electrical measurements are performed on three test structures: BPR, single nTSV, and BPR-TSV connected chains. Some design variations (e.g., CD and barrier thickness) are also included. The BPR/nTSV resistance

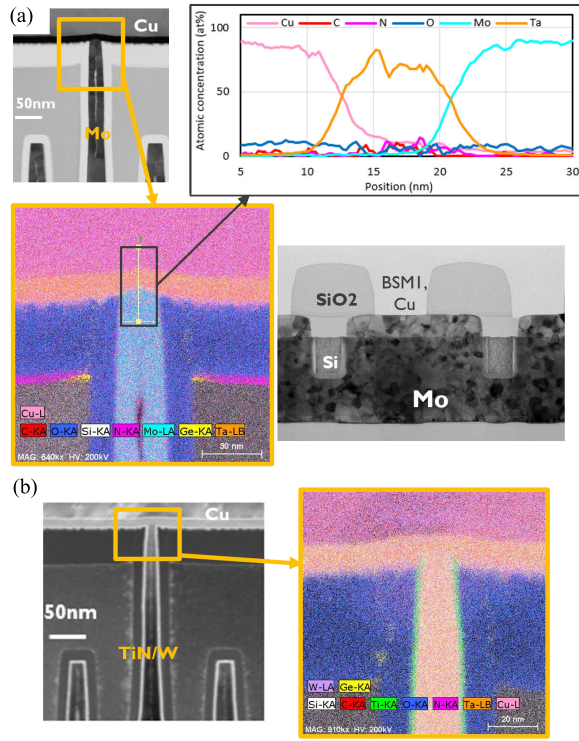


Fig. 7. TEM image and EDS analysis at the interface between Cu BSM1 and (a) Mo-filled vias and (b) TiN/W-filled vias.

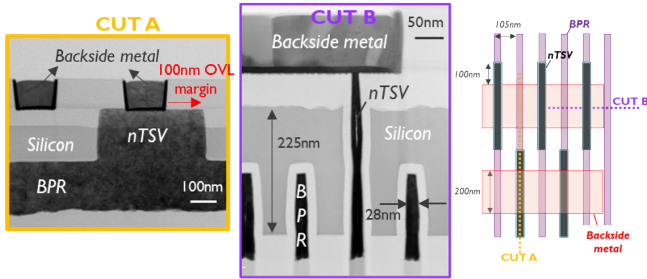


Fig. 8. TEM images of FS-BS connections consisting of TiN/W-filled BPR/nTSV and Cu BSM1 in two orthogonal directions, along with an illustrative schematic. Note the nTSV profile is not final.

has a direct impact on the BSPDN IR-drop [15], favoring low-resistivity metals such as Mo over TiN/W. Compatibility of barrier-free Mo with FEOL processing was demonstrated for the BPR [16]. For the first time, Mo-filled nTSVs are demonstrated for FS-BS connection and compared to TiN/W-filled counterparts using the FS nTSV flow.

A. BPR Resistance

The BPR resistance is extracted using the test structure shown in the inset of Fig. 9(a). BPR length varies from 2, 10–100 μm . The normalized resistances (i.e., resistance per unit length) of Mo and TiN/W filling are compared. It is observed that Mo-filled wafers have two to three times lower BPR resistance than the TiN/W-filled counterparts. Meanwhile, the tight distribution of Mo wafers also indicates a stable metallization process. In comparison, the dispersed distribution of TiN/W wafers can be largely attributed to the within-wafer barrier thickness variation. Additionally,

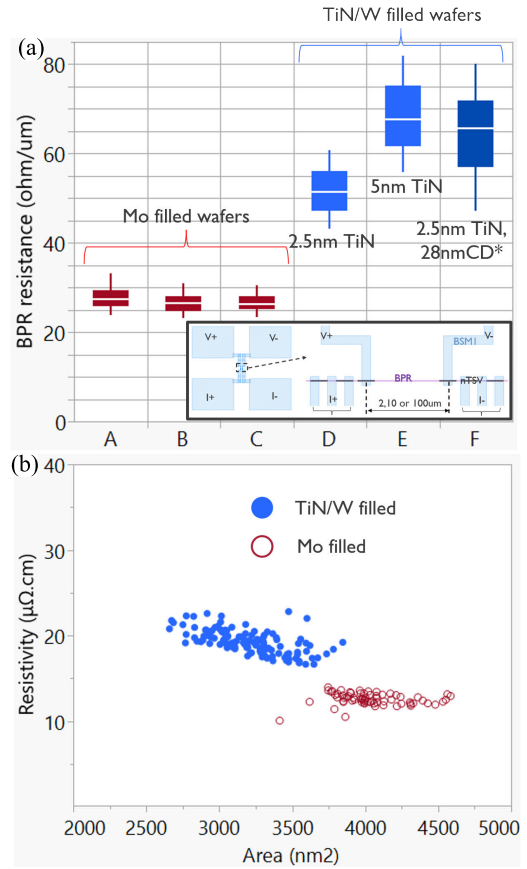


Fig. 9. (a) BPR resistance comparison of Mo-filled wafers and TiN/W-filled wafers. (b) Mo and TiN/W resistivity as a function of area, from TCR test.

the interwafer CD/barrier thickness-dependent BPR resistance trend is as expected, since either 28 nm CD (instead of 36 nm) or 5 nm TiN (instead of 2.5 nm) will result in a smaller W volume.

Three wafers (two TiN/W wafers and one Mo wafer) are sent for temperature-coefficient-of-resistivity (TCR) measurement to extract the resistivity and cross-sectional area [17]. As shown in Fig. 9(b), the resistivity of Mo is approximately 30% lower than that of TiN/W.

B. nTSV Kelvin Resistance

Similarly, the comparison of single nTSV Kelvin resistance is shown in Fig. 10(a). The test structure used is illustrated in the inset. The average nTSV resistance of three Mo-filled wafers is 5.3 Ω . A likewise tight distribution is observed. However, in terms of TiN/W filled wafers, the nTSV resistance can reach up to 15 Ω (outliers have even higher values). The large distribution range is not only related to the TiN barrier thickness but also the TSV etching profile.

In this respect, a sensitivity simulation of the TiN/W nTSV resistance is performed, considering variations in TiN barrier thickness and the TSV slope angle. Taking process relevance into account, the barrier thickness range is restricted from 1.5 to 5 nm. The TSV slope angle is also restricted to a range of 87°–89° based on the experimental observations.

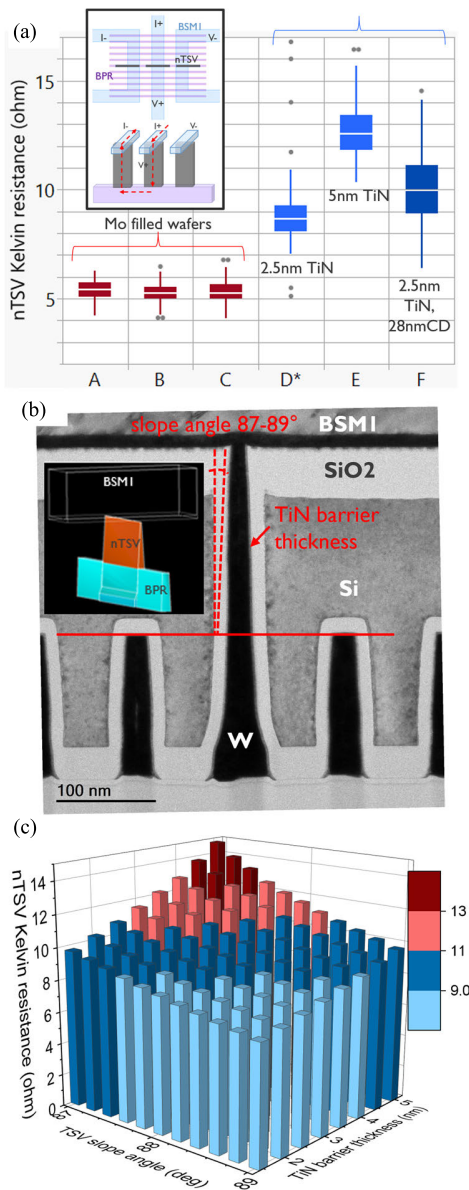


Fig. 10. (a) Single nTSV Kelvin resistance comparison of Mo-filled wafers and TiN/W filled wafers. (b) Three-dimensional model built for simulation, with dimensions calibrated from TEM image. (c) Sensitivity plot of nTSV Kelvin resistance as a function of TSV slope angle and TiN barrier thickness.

A 3-D model has been developed, with dimensions carefully calibrated using TEM images [Fig. 10(b)] and electrical parameters aligned with the process database. The simulated results, displayed in Fig. 10(c), qualitatively align with the trends in the measured data. In particular, by assuming a $\sim 88^\circ$ etch slope angle, the TiN thickness values of 2.5 and 5 nm result in simulated resistances of 9.1 and 11.6 Ω , respectively, in good agreement with the median resistance values measured on wafers D and E, respectively, as shown in Fig. 10(a). These results indicate that the TiN/W nTSV resistance can be reduced to less than 10 Ω , given a straight etching profile and an extremely thin barrier thickness. However, the increasing reliability concerns still need to be evaluated.

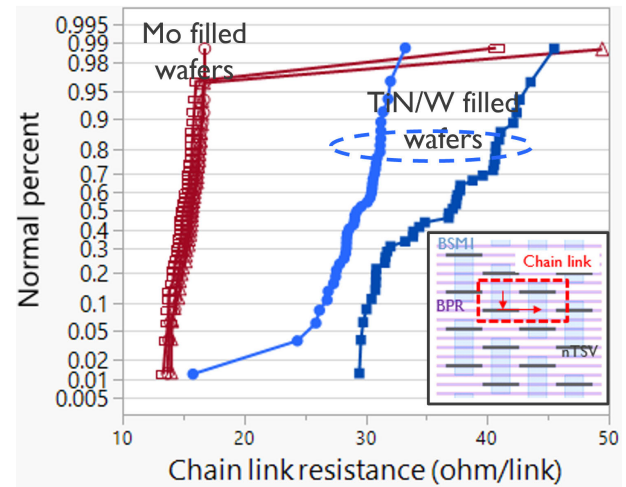


Fig. 11. Cumulative plot of unit chain link resistance.

C. Chain Link Resistance

As the inset of Fig. 11 indicates, a long chain link resistance is the sum of a certain number of unit chain resistance. The unit chain resistance comes from two parts: the resistance of a single nTSV and the resistance of the corresponding BPR segment, which are connected in series (the contribution from the BSM1 segment is negligible due to its large CD). Therefore, the unit chain resistance can be expressed by the equation

$$R_{\text{chain}} \approx R_{\text{BPR}} \times L_{\text{BPR}} + R_{\text{nTSV}} \quad (1)$$

where L_{BPR} is the BPR segment length. Chains with different unit numbers are measured (from 60 to 3800). Fig. 11 shows the cumulative plot of chain link resistance after normalization (i.e., resistance per unit chain). As expected, the comparisons between Mo and TiN/W filled wafers are consistent with the previous discussions in the BPR and nTSV sections.

In the future, implementing a more detailed process monitoring system that focuses on process variability at different fabrication steps, and correlating this variability with electrical performance, could be highly beneficial. In particular, when integrating the BPR-nTSV first scheme with active devices, it will be crucial to identify process-related factors that could impact the final performance of the active devices. Meanwhile, the continuous exploration of flexible thinning sequences and alternative metal filling options remains of high significance. Additionally, the reliability assessment of different metal-filled vias is also essential.

V. CONCLUSION

By patterning self-aligned slit nTSV to BPR during FS processing, the overlay requirement for the BS metal routing is significantly relaxed, eliminating the need for HOCPE during BS patterning. Due to the improvements in TTV using a revised extreme wafer thinning sequence, the thickness of $\text{Si}_{0.75}\text{Ge}_{0.25}$ ESL is reduced from 50 to 10 nm. In terms of electrical performance, Mo-filled wafers exhibit lower resistance and tighter data distribution across various test structures as

compared to TiN/W-filled counterparts. In brief, an alternative nTSV to BPR connection scheme defined from wafer FS is introduced, enriching the toolbox for implementing BSPDN in FinFETs, nanosheets, or CFETs.

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